

Lab Goal : This lab was designed to teach you more about Heaps.

Lab Description : Create a class of your choosing that stores some data you cook up. The class must implement Comparable as it will be stored in a min heap. Instantiate and store at least 8 Objects of the type you create. Print out the 8 Objects in ascending order. A Monster class has been provided, or you can use another type of object that implements the Comparable interface, just not a String or Wrapper class such as Integer, Double, etc.

Sample Data and Output (this output would apply if you were storing Monster):

```
Heap<String> test = new Heap<String>();
test.add("a");
test.add("5");
test.add("B");
test.add("15");
test.add("1A");
test.add("a2");
testIt(test);
```

```
Heap<Monster> test2 = new Heap<Monster>();
test2.add(new Monster(1,5));
test2.add(new Monster(1,3));
test2.add(new Monster(10,55));
test2.add(new Monster(12,45));
test2.add(new Monster(111,123));
test2.add(new Monster(5,9));
testIt(test2);
```

Example2 - The Heap of Strings:

PRINTING THE HEAP!

```
    a2
  5   a
15  1A  B
```

```
The heap as an array: [a2, 5, a, 15, 1A, B]
peek ----> a2
remove --> a2
```

PRINTING THE HEAP!

```
    a
  5   B
15  1A
```

```
The heap as an array: [a, 5, B, 15, 1A]
peek ----> a
remove --> a
```

PRINTING THE HEAP!

```
    B
  5  1A
15
```

```
public static void testIt(Heap heap) {
    heap.print();
    System.out.println("The heap as an array: " + heap);
    System.out.println("peek ----> " + heap.peek());
    System.out.println("remove --> " + heap.remove());
    heap.print();
    System.out.println("The heap as an array: " + heap);
    System.out.println("peek ----> " + heap.peek());
    System.out.println("remove --> " + heap.remove());
    heap.print();
}
```

Example2 - The Heap of Monsters:

PRINTING THE HEAP!

```
    111-123
  12-45  5-9
1-3  10-55  1-5
```

The heap as an array: [111-123, 12-45, 5-9, 1-3, 10-55, 1-5]

peek ----> 111-123

remove --> 111-123

PRINTING THE HEAP!

```
    12-45
  10-55  5-9
1-3  1-5
```

The heap as an array: [12-45, 10-55, 5-9, 1-3, 1-5]

peek ----> 12-45

remove --> 12-45

PRINTING THE HEAP!

```
    10-55
  1-5  5-9
1-3
```